Assignment 5: Database Creation and Table Implementation

DBMS Assignment Series | Project topic: **Online Food Delivery System (QuickBite)**

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1. Objective

Create the database for the proposed Online Food Delivery System and implement every table from the finalized design (Assignment 3) using SQL DDL commands, with all relationships defined through primary and foreign keys.

2. Database creation

```
CREATE DATABASE quickbite_db;   -- MySQL / PostgreSQL
USE quickbite_db;                  -- MySQL
-- SQLite: the database is the file itself:  sqlite3 quickbite_db.sqlite
```

Explanation: `CREATE DATABASE` makes a new, empty container named **quickbite_db** that will hold all six tables of the system.

Note: All SQL below was executed on SQLite 3.45 and the outputs are real. Syntax is standard SQL; if your course uses MySQL, change `AUTOINCREMENT` to `AUTO_INCREMENT`. `CURRENT_DATE` defaults work as-is on MySQL 8.

3. Table creation (DDL)

Tables are created in dependency order: parent tables (*customers, restaurants, delivery_partners*) first, then *menu_items*, *orders* and finally *order_items*, which reference them. Otherwise a foreign key would point to a table that does not exist yet.

```
CREATE TABLE customers (
    customer_id  INTEGER PRIMARY KEY AUTOINCREMENT,
    full_name    VARCHAR(100) NOT NULL,
    email        VARCHAR(100) NOT NULL UNIQUE CHECK (email LIKE '%_@_%._%'),
    phone        CHAR(10) NOT NULL UNIQUE CHECK (LENGTH(phone) = 10),
    area         VARCHAR(50) NOT NULL DEFAULT 'Koramangala',
    signup_date  DATE NOT NULL DEFAULT CURRENT_DATE,
    membership   VARCHAR(10) NOT NULL DEFAULT 'Regular'
                 CHECK (membership IN ('Regular', 'Premium'))
);

CREATE TABLE restaurants (
    restaurant_id    INTEGER PRIMARY KEY AUTOINCREMENT,
    restaurant_name  VARCHAR(100) NOT NULL,
    cuisine          VARCHAR(40) NOT NULL,
    area             VARCHAR(50) NOT NULL,
    rating           DECIMAL(2,1) NOT NULL DEFAULT 0 CHECK (rating BETWEEN 0 AND 5),
    is_open          INTEGER NOT NULL DEFAULT 1 CHECK (is_open IN (0, 1)),
    UNIQUE (restaurant_name, area)
);

CREATE TABLE delivery_partners (
    partner_id    INTEGER PRIMARY KEY AUTOINCREMENT,
    partner_name  VARCHAR(100) NOT NULL,
    phone         CHAR(10) NOT NULL UNIQUE CHECK (LENGTH(phone) = 10),
    vehicle       VARCHAR(10) NOT NULL DEFAULT 'Bike'
                  CHECK (vehicle IN ('Bike', 'Scooter', 'Cycle')),
    availability  VARCHAR(10) NOT NULL DEFAULT 'Available'
                  CHECK (availability IN ('Available', 'Busy', 'Offline'))
);

CREATE TABLE menu_items (
    item_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    restaurant_id  INTEGER NOT NULL,
    item_name      VARCHAR(100) NOT NULL,
    category       VARCHAR(15) NOT NULL
                   CHECK (category IN ('Starter', 'Main Course', 'Dessert', 'Beverage')),
    price          DECIMAL(7,2) NOT NULL CHECK (price > 0),
    is_veg         INTEGER NOT NULL DEFAULT 1 CHECK (is_veg IN (0, 1)),
    UNIQUE (restaurant_id, item_name),
```

```
    FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);

CREATE TABLE orders (
    order_id        INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id     INTEGER NOT NULL,
    restaurant_id   INTEGER NOT NULL,
    partner_id      INTEGER,
    order_date      DATE NOT NULL DEFAULT CURRENT_DATE,
    status          VARCHAR(10) NOT NULL DEFAULT 'Placed'
                    CHECK (status IN ('Placed', 'Delivered', 'Cancelled')),
    payment_method  VARCHAR(10) NOT NULL DEFAULT 'UPI'
                    CHECK (payment_method IN ('UPI', 'Card', 'Cash')),
    delivery_fee    DECIMAL(6,2) NOT NULL DEFAULT 30 CHECK (delivery_fee >= 0),
    FOREIGN KEY (customer_id)   REFERENCES customers (customer_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    FOREIGN KEY (restaurant_id) REFERENCES restaurants (restaurant_id)
        ON UPDATE CASCADE ON DELETE RESTRICT,
    FOREIGN KEY (partner_id)    REFERENCES delivery_partners (partner_id)
        ON UPDATE CASCADE ON DELETE SET NULL
);

CREATE TABLE order_items (
    order_item_id  INTEGER PRIMARY KEY AUTOINCREMENT,
    order_id       INTEGER NOT NULL,
    item_id        INTEGER NOT NULL,
    quantity       INTEGER NOT NULL DEFAULT 1 CHECK (quantity BETWEEN 1 AND 20),
    unit_price     DECIMAL(7,2) NOT NULL CHECK (unit_price > 0),
    UNIQUE (order_id, item_id),
    FOREIGN KEY (order_id) REFERENCES orders (order_id)
        ON UPDATE CASCADE ON DELETE CASCADE,
    FOREIGN KEY (item_id)  REFERENCES menu_items (item_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
);
```

4. Verification of the created structure

4.1 Tables present in the database

```
SELECT name FROM sqlite_master WHERE type = 'table' AND name NOT LIKE 'sqlite_%';
```

table_name
customers
restaurants
delivery_partners
menu_items
orders
order_items

4.2 Structure of every table (equivalent of DESCRIBE)

```
PRAGMA table_info(<table_name>);   -- MySQL: DESCRIBE <table_name>;
```

table	column	data type	NOT NULL	default	key
customers	customer_id	INTEGER	YES	-	PK
customers	full_name	VARCHAR(100)	YES	-	
customers	email	VARCHAR(100)	YES	-	
customers	phone	CHAR(10)	YES	-	
customers	area	VARCHAR(50)	YES	'Koramangala'	
customers	signup_date	DATE	YES	CURRENT_DATE	
customers	membership	VARCHAR(10)	YES	'Regular'	
restaurants	restaurant_id	INTEGER	YES	-	PK
restaurants	restaurant_name	VARCHAR(100)	YES	-	
restaurants	cuisine	VARCHAR(40)	YES	-	
restaurants	area	VARCHAR(50)	YES	-	

table	column	data type	NOT NULL	default	key
restaurants	rating	DECIMAL(2,1)	YES	0	
restaurants	is_open	INTEGER	YES	1	
delivery_partners	partner_id	INTEGER	YES	-	PK
delivery_partners	partner_name	VARCHAR(100)	YES	-	
delivery_partners	phone	CHAR(10)	YES	-	
delivery_partners	vehicle	VARCHAR(10)	YES	'Bike'	
delivery_partners	availability	VARCHAR(10)	YES	'Available'	
menu_items	item_id	INTEGER	YES	-	PK
menu_items	restaurant_id	INTEGER	YES	-	
menu_items	item_name	VARCHAR(100)	YES	-	
menu_items	category	VARCHAR(15)	YES	-	
menu_items	price	DECIMAL(7,2)	YES	-	
menu_items	is_veg	INTEGER	YES	1	
orders	order_id	INTEGER	YES	-	PK
orders	customer_id	INTEGER	YES	-	
orders	restaurant_id	INTEGER	YES	-	
orders	partner_id	INTEGER	no	-	
orders	order_date	DATE	YES	CURRENT_DATE	
orders	status	VARCHAR(10)	YES	'Placed'	
orders	payment_method	VARCHAR(10)	YES	'UPI'	
orders	delivery_fee	DECIMAL(6,2)	YES	30	
order_items	order_item_id	INTEGER	YES	-	PK
order_items	order_id	INTEGER	YES	-	
order_items	item_id	INTEGER	YES	-	
order_items	quantity	INTEGER	YES	1	
order_items	unit_price	DECIMAL(7,2)	YES	-	

4.3 Relationships (foreign keys)

```
PRAGMA foreign_key_list(<table_name>);   -- MySQL: SHOW CREATE TABLE <table_name>;
```

child table	FK column	references	ON DELETE	ON UPDATE
menu_items	restaurant_id	restaurants(restaurant_id)	RESTRICT	CASCADE
orders	partner_id	delivery_partners(partner_id)	SET NULL	CASCADE
orders	restaurant_id	restaurants(restaurant_id)	RESTRICT	CASCADE
orders	customer_id	customers(customer_id)	RESTRICT	CASCADE
order_items	item_id	menu_items(item_id)	RESTRICT	CASCADE
order_items	order_id	orders(order_id)	CASCADE	CASCADE

5. Relationships summary

- **customers 1 : N orders** - a customer places many orders; each order belongs to one customer.
- **restaurants 1 : N menu_items** and **restaurants 1 : N orders** - a restaurant offers many dishes and receives many orders.
- **delivery_partners 1 : N orders** - a partner delivers many orders; partner_id may be NULL until a partner is assigned (and is set to NULL if a partner leaves).
- **orders M : N menu_items** - resolved by the junction table *order_items*, which stores quantity and the unit price at the time of ordering.

All tables were created without errors and the foreign keys above confirm that the relationships are properly defined.